DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Physics for Computer Science Cluster | Course Code: 25EN1115 | Credits: 4

Exam Session: End Semester Exam May 2026 (Max Marks: 80, Duration: 2h 30m) Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Set up the 1D time-independent Schrödinger wave equation for a particle trapped in an infinite potential well of width L. Solve for the normalized wave functions and discrete energy eigenvalues En = (n^2 * h^2) / (8 * m * L^2). **[8 Marks]**

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Q2. Define a Qubit and explain Quantum Entanglement. Write the matrix representation of Pauli-X gate and show its action on basis states |0> and |1>. Describe the working of Quantum Key Distribution (QKD). **[8 Marks]**

MODULE 2

Q3. Explain the principle, energy level transitions, and working of an Nd:YAG 4-level solid-state laser with a labelled diagram. **[8 Marks]**

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Q4. Derive an expression for Hall Voltage (VH) in a conducting semiconductor strip carrying current I in a transverse magnetic field B. Explain how Hall Effect determines majority carrier type and concentration. **[8 Marks]**